

1. Let  $X \sim B(30, 0.3)$ . Find:
  - a.  $P(X = 10)$
  - b.  $P(X \leq 7)$
  - c.  $P(X < 7)$
  - d.  $P(X \geq 10)$
  - e.  $P(4 \leq X \leq 12)$
2. A gambler plays roulette, and always puts one pound on 20 black. The probability of winning on a single spin of the wheel is  $\frac{1}{37}$ . If the gambler tries his luck for 50 spins, then what is the probability that he gets:
  - a. at least one win;
  - b. exactly one win;
  - c. more than two wins?
3. The probability that a particular Fifth Form boy will forget his calculator for his maths lesson is 0.35, independently of any other time. In sixteen lessons, what is the probability that this boy will forget his calculator
  - a. on exactly 6 occasions;
  - b. on fewer than 4 occasions;
  - c. more than half the time?
4. According to the manufacturer, one quarter of a brand of gummy sweets are red. If you purchase a bag containing 32 of these sweets, then what is the probability that the number of red sweets is
  - a. exactly 8;
  - b. at least 10;
  - c. fewer than 6;
  - d. no more than 9;
  - e. between 5 and 15 inclusive.
5. Han-Xi flips a fair coin 18 times. Given that more than half of her flips come up heads, what is the probability that fewer than 14 of her flips are heads?
6. In a game, you roll nine (fair) six-sided dice, and flip three coins. You score one point for every 6 you roll on the dice, and one point for every head you flip.
  - a. What is the probability that you score exactly four points?
  - b. Given that you scored exactly four points, what is the probability that exactly two of the coins showed heads?
7. Every day you meet Mrs Truddlewump in her office, and she randomly offers you cake with probability  $\frac{1}{3}$ . With the aid of tables, compute the probability that you find yourself eating cake on the seventeenth day for at least the sixth time.
8. A fireworks manufacturer makes rockets that fire correctly 90% of the time. They sell their rockets in packs of twelve.
  - a. If Charlie purchases two packs, then what is the probability that exactly 20 of the rockets fire correctly?
  - b. If Rachel purchases one pack, what is the probability that less than three quarters of the rockets fire correctly.
  - c. Each pack comes with the promise: "Guaranteed that at least  $N$  rockets will fire correctly, or your money back". If the company wants to give the money back no more than 5% of the time, then what number should they choose for  $N$ ?

9. A company manufactures chocolate eggs which contain a surprise toy. This year, the toys come in three categories – fairies, dinosaurs, and cars – with an approximately equal chance of receiving each.
  - a. If Joe's parents purchase 25 such eggs for an Easter party, then what is the probability that exactly 10 of them contain fairies?
  - b. If Han-Xi purchases a dozen of these eggs for her class, what is the probability that at least one third, but less than two-thirds, of the eggs contain cars?
  - c. Padma's parents want to buy her some of these eggs for her fifth birthday. What is the minimum number of eggs should they buy so that Padma has at least a 90% chance of getting at least one dinosaur toy.
10. A gardener wishes grows a particular type of plant from seeds, and believes that the probability that a seed will germinate and grow into a healthy plant is 0.7. The gardener wants ten such plants (and more than ten plants isn't a problem).
  - a. If the gardener plants a pack of 14 seeds, then what is the probability that he achieves his aim?
  - b. What is the minimum number of seeds that the gardener should plant to ensure that there is at least a 95% chance of fulfilling his aim?
11. In the make-believe land of Costador 5% of one-Nuyen coins are fake. If you want to purchase a coffee costing 4 Nuyen, then how many coins should you bring with you to ensure that you have a 99% chance of satisfying your caffeine needs?
12. In a game, you roll up to 20 six-sided dice to determine your available actions for the turn. The more dice you roll, the more things you'll be able to do (and you'll have a greater choice of different actions too). One symbol on the dice is bad, however, and if you roll three or more of these symbols you lose your turn!  
If you are prepared to miss up to one in every four of your turns, on average, then what is the maximum number of dice you should roll each turn?
13. A jury consists of 12 jurors. Each juror makes the correct decision with probability 0.8, and a defendant is only declared to be guilty if more than half of the jurors reach that decision.  
What is the probability that:
  - a. a guilty defendant is set free;
  - b. an innocent defendant is sent down?

Now suppose that 90%, on average, of defendants brought before the court are guilty.

  - c. Given that the jury reaches a verdict of 'guilty' what is the probability that the defendant is innocent?
  - d. Given that the jury reaches a verdict of 'not-guilty' what is the probability that the defendant is guilty?
14. A childcare company has 20 children registered at one of their nurseries, and is required to provide supervision on a two-to-one basis (so they need one adult present on site for every two children present). If they don't have enough staff on site to supervise the children, they have to bring in extra personnel at a premium price. Based on their experience, the company believes that each child attends nursery roughly half of the time. The nursery operates 248 days of the year. What is the minimum number of staff that the company need to ensure that on average there will be no more than five days a year in which they need to bring in extra personnel?
15. A bag contains 2 red balls, 3 blue balls, and 5 white balls. Ten balls are sampled from the bag with replacement. Find the probability that:
  - a. there are 5 white balls and 5 blue balls (and no red balls);
  - b. there are 2 red balls, 3 blue balls, and 5 white balls;
  - c. there are 3 red balls, 4 blue balls, and 3 white balls;
  - d. exactly 4 of the balls are white;
  - e. all the balls are blue and white;
  - f. at least 5 of the balls are red or blue;
  - g. there are at least 3 red balls, given that there are exactly 5 white balls.